
Mechanics

Bionics Eng.

Lecture 17

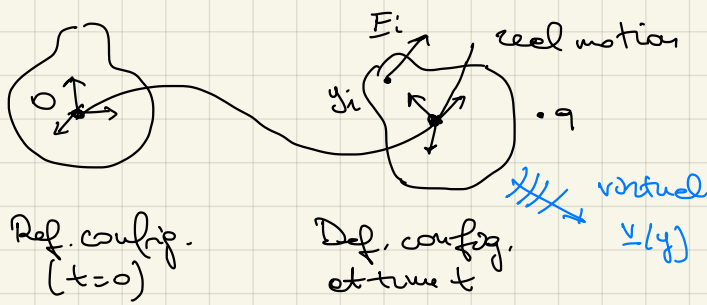
Complements on
AP V - 1

1/xii/23



Principle of virtual ^{work} power (PVP)

Recall the version for rigid bodies



$$\underline{F} = \sum \underline{F}_i = 0$$

$$\underline{M}_q = \sum (y_i - q) \wedge \underline{F}_i = 0$$

bodies $\underline{F}_i, i=1, \dots, n$

$$\left. \begin{aligned} \underline{F} = \sum \underline{F}_i = 0 \\ \underline{M}_q = \sum (y_i - q) \wedge \underline{F}_i = 0 \end{aligned} \right\} \Leftrightarrow P_{\text{ext}}[\underline{L}](\underline{v}) = 0$$

$\forall \underline{v}$ virtual rigid velocity field = admissible test field

= virtual velocity field or "test" velocity field

$\underline{v}(y)$ = field of velocities in a virtual rigid motion away from current / deformed configuration.
 = $\underline{v}_q + \underline{\omega} \wedge (y - q)$ in which $\underline{v}_q$ and $\underline{\omega}$ are arbitrary

i.e., let forces expend power against it

$$\sum \underline{F}_i \cdot \underline{v}(y_i) = \sum \underline{F}_i \cdot (\underline{v}_q + \underline{\omega} \wedge (y_i - q)) = \underbrace{\underline{v}_q \cdot \sum \underline{F}_i}_{\underline{F}} + \underbrace{\underline{\omega} \cdot \sum (y_i - q) \wedge \underline{F}_i}_{\underline{M}_q} = 0 \quad \forall \underline{v}_q, \underline{\omega} \Leftrightarrow \underline{F} = 0, \underline{M}_q = 0$$

(in particular, in 2D,
 $\underline{\omega} = \omega \underline{e}_3, \underline{M}_q = M_q \underline{e}_3$)

Idea: instead of saying that something = 0, one says that it's orthogonal to every test vector, i.e., its scalar product with every test vector is = 0.

test vector = virtual velocity

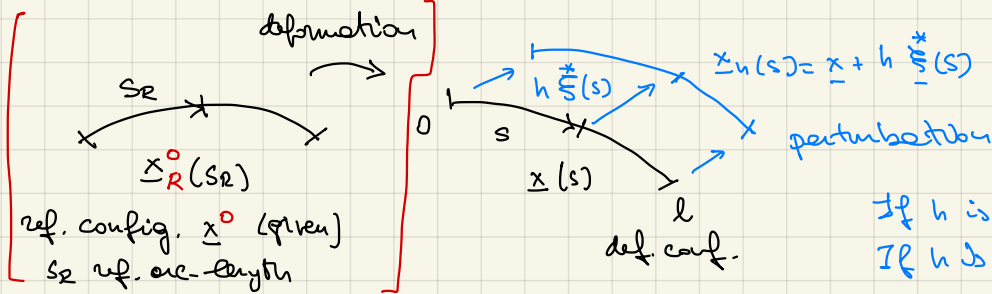
scalar product = power expended

One can generalize this idea to force densities and virtual velocity fields

We'll get in this way a variational formulation of the equilibrium equations, based on computing the power expended against virtual variations of the current configuration.

Executive summary for PPN/PLV

Kinematics of virtual displacement fields for planar rods



If h is time, $\underline{\bar{x}}^*$ is velocity (virtual)
If h is non-dim., $\underline{\bar{x}}^*$ is displacement (virtual)

Any quantity G [say, gravitational energy] depending on $\underline{x}$ becomes a function G_h of h through $\underline{x}_h$

$$G_h = G(\underline{x}_h) = G(\underline{x} + h \underline{\bar{x}}^*)$$

Consider $\frac{d}{dh} G_h \Big|_{h=0} =: \bar{G} = \delta G$ (It's the "directional derivative" along $\underline{\bar{x}}^*$)

$$[e.g. G_h = G(\underline{x}_h) = \int_0^l \rho(s) g \underbrace{\underline{x}_{h2}(s)}_{\underline{x}_2(s) + h \underline{\bar{x}}_2^*(s)} ds = \int_0^l \rho g \underline{x}_2 ds + h \underbrace{\int_0^l \rho g \underline{\bar{x}}_2^* ds}_{= \bar{G}}]$$

Quantities of interest: $\underline{t}_h, \underline{n}_h, \alpha_h, s_h$ from $\underline{s}'_h = |\underline{x}'_h|$, $c_h = \frac{\alpha'_h}{s'_h}$ (geometry)

$$\underline{\bar{s}}_h, u_h = \underline{\bar{s}}_h \cdot \underline{t}, v_h = \underline{\bar{s}}_h \cdot \underline{n}$$

$$\omega_h = \alpha_h - \alpha, \epsilon_h = s'_h - 1, k_h = \omega'_h$$

} deformation from $\underline{x}$ to $\underline{x}_h$

For each of these quantities can compute $\frac{d}{dh} G_h = \bar{G}$. After some calculations

$$\left[\underline{\bar{s}}_h^* = \underline{\bar{u}} \underline{t} + \underline{\bar{w}} \underline{n}, \underline{\bar{s}}_h'^* = \underline{\bar{\epsilon}} \underline{t} + \underline{\bar{\omega}} \underline{n}, \underline{\bar{k}} = \underline{\bar{\omega}}' \right]$$

key equations
(kinematic compatibility linking virtual strains $\underline{\bar{\epsilon}}, \underline{\bar{k}}$ with virtual displacements $\underline{\bar{u}}, \underline{\bar{w}}, \underline{\bar{\omega}}$)

$$[|s'|^* = (1 + \epsilon)^* = \epsilon^*] \quad \text{skip}$$

Then are reminiscent of linear kinematics for straight rods $\underline{t}' = 0, \underline{n}' = 0$

$$\underline{\bar{s}}_h^* = \underline{\bar{u}} \underline{t} + \underline{\bar{w}} \underline{n}, \underline{\bar{s}}_h'^* = \underline{\bar{u}}' \underline{t} + \underline{\bar{w}}' \underline{n} = \underline{\bar{\epsilon}} \underline{t} + \underline{\bar{\omega}} \underline{n}, \underline{\bar{k}} = \underline{\bar{\omega}}'$$

$$\underline{\bar{\epsilon}} = \underline{\bar{u}}', \underline{\bar{\omega}} = \underline{\bar{w}}', \underline{\bar{k}} = \underline{\bar{\omega}}''$$

The deformed configuration $\underline{x}(s)$ can be thought of as the reference configuration for the virtual deformation that produces $\underline{x}_h$.

For $\underline{x}_h$, can define the usual quantities $\rightarrow$ For each quantity G_h , can compute $\dot{G}^* = \frac{d}{dh} G_h \Big|_{h=0}$ (see next pages for calculations)

$$\underline{t}_h$$

$$\underline{n}_h = \underline{e}_3 \wedge \underline{t}_h$$

$$\alpha_h \text{ from } \begin{cases} \cos \alpha_h = \underline{t}_h \cdot \underline{e}_1 \\ \sin \alpha_h = \underline{t}_h \cdot \underline{e}_2 \end{cases}$$

$$s_h \text{ from } s'_h = |\underline{x}'_h|$$

$$C_h = \frac{\alpha'_h}{s'_h}$$

$$\underline{t}^* = \underline{\underline{s}}'^* - (\underline{t} \cdot \underline{\underline{s}}'^*) \underline{t} = (\underline{I}_2 - \underline{t} \otimes \underline{t}) \underline{\underline{s}}'^* \quad (3)$$

$$(\underline{s}')^* = \underline{\underline{s}}'^* \cdot \underline{t} \quad (1)$$

$$\underline{\underline{s}}'_h = \underline{x}_h - \underline{x}$$

$$u_h = \underline{\underline{s}}'_h \cdot \underline{t}$$

$$w_h = \underline{\underline{s}}'_h \cdot \underline{n}$$

$$\omega_h = \alpha_h - \alpha \text{ from } \begin{cases} \cos \omega_h = \underline{t}_h \cdot \underline{t} \\ \sin \omega_h = \underline{t}_h \cdot \underline{n} \end{cases}$$

$$E_h = s'_h - 1$$

$$\chi_h = \omega'_h$$

$$\underline{\underline{s}}'^* = \underline{\underline{s}}'^*$$

$$\underline{\underline{e}}^* = \underline{\underline{s}}'^* \cdot \underline{t}$$

$$\underline{\underline{w}}^* = \underline{\underline{s}}'^* \cdot \underline{n}$$

$$\underline{\underline{\omega}}^* = \underline{\underline{s}}'^* \cdot \underline{n} \quad (4)$$

$$\underline{\underline{e}}^* = \underline{\underline{s}}'^* \cdot \underline{t} \quad (5)$$

$$\underline{\underline{\chi}}^* = \underline{\underline{\omega}}^* \quad (7)$$

(*)

Remark $\omega_h = \alpha_h - \alpha$ is the angle between $\underline{t}_h$ and $\underline{t}$, hence $\cos \omega_h = \underline{t}_h \cdot \underline{t}$

Remark (*) is $\underline{\underline{s}}'^* = \underline{\underline{e}}^* \underline{t} + \underline{\underline{w}}^* \underline{n}$. Equations (4) and (5) give

$$\underline{\underline{s}}'^* = \underline{\underline{e}}^* \underline{t} + \underline{\underline{w}}^* \underline{n}$$

and (7) is $\underline{\underline{\chi}}^* = \underline{\underline{w}}^*$

key equations for later.

These equations are reminiscent of those of linearized kinematics

$$\underline{\underline{e}}^* = \underline{\underline{u}}', \quad \underline{\underline{\omega}}^* = \underline{\underline{w}}', \quad \underline{\underline{\chi}}^* = \underline{\underline{w}}''$$

Geometry of $\underline{x}_h$

Deformation from $\underline{x}$ to $\underline{x}_h$

$$1. (S')^* = \frac{d}{dh} |x'_h| \Big|_{h=0} = ?$$

[we use: $(\underline{v} \cdot \underline{v})' = 2 \underline{v} \cdot \underline{v}'$]

$$\begin{aligned} \frac{d}{dh} |x'_h| \Big|_{h=0} &= \frac{d}{dh} \left((x'_h + h \underline{\xi}'^*) \cdot (x'_h + h \underline{\xi}'^*) \right)^{1/2} \Big|_{h=0} = \frac{1}{2} \left((x'_h + h \underline{\xi}'^*) \cdot (x'_h + h \underline{\xi}'^*) \right)^{-1/2} \cdot \frac{d}{dh} (x'_h + h \underline{\xi}'^*) \Big|_{h=0} \\ &= \underbrace{|x'|^{-1}}_{=1} \cdot \underbrace{x'}_{=t} \cdot \underbrace{\underline{\xi}'^*}_{= \underline{\xi}'^* \cdot t} = \underline{\xi}'^* \cdot t \end{aligned}$$

because s arc-length of $\underline{x}(s)$

$$2. \frac{d}{dh} \frac{1}{|x'_h|} \Big|_{h=0} = \frac{d}{dh} |x'_h|^{-1} \Big|_{h=0} = -1 |x'_h|^{-2} \frac{d}{dh} |x'_h| \Big|_{h=0} = - \underbrace{|x'|^{-2}}_{=1} \underline{\xi}'^* \cdot t \quad (\text{using 1.}) = - \underline{\xi}'^* \cdot t$$

$$3. \frac{d}{dh} \frac{x'_h}{|x'_h|} \Big|_{h=0} = \frac{d}{dh} \underline{t}_h \Big|_{h=0} = \underline{t}^* = ?$$

$$\begin{aligned} \frac{d}{dh} \left(\frac{1}{|x'_h|} x'_h \right) \Big|_{h=0} &= \frac{d}{dh} \frac{1}{|x'_h|} \Big|_{h=0} x'_h \Big|_{h=0} + \frac{1}{|x'_h|} \Big|_{h=0} \frac{d}{dh} (x'_h + h \underline{\xi}'^*) \Big|_{h=0} \quad (\text{using 2}) \\ &= \underbrace{- \underline{\xi}'^* \cdot t}_{= \underline{t}^*} \cdot \underbrace{x'}_{=t} + \frac{1}{\underbrace{|x'|}_{=1}} \underline{\xi}'^* = \underline{\xi}'^* - (\underline{\xi}'^* \cdot t) t = (\underline{I}_2 - t \otimes t) \underline{\xi}'^* \end{aligned}$$

$$4. \sin \omega_h = \underline{n} \cdot \underline{t}_h = \underline{n} \cdot \frac{x'_h}{|x'_h|}$$

$$\begin{aligned} \frac{d}{dh} \sin \omega_h \Big|_{h=0} &= \cos \omega_h \frac{d}{dh} \omega_h \Big|_{h=0} = 1 \cdot \underline{\omega}^* = \underline{n} \cdot \frac{d}{dh} \frac{x'_h}{|x'_h|} \Big|_{h=0} = \underline{n} \cdot \underline{t}^* = \underline{n} \cdot \underline{\xi}'^* \\ \omega_h = \alpha_h - \alpha = 0 \text{ for } h=0 &\quad \cos \omega_h \Big|_{h=0} = 1 \end{aligned}$$

$\Rightarrow \underline{\omega}^* = \underline{\xi}'^* \cdot \underline{n}$

$$5. 1 + \varepsilon_h = s'_h = |x'_h| \Rightarrow \frac{d}{dh} (1 + \varepsilon_h) \Big|_{h=0} = \underline{\varepsilon}^* = (s')^* = \underline{\xi}'^* \cdot t \quad (\text{using 1})$$

Combining 5. and 4. we get $\underline{\xi}'^* = (\underline{\xi}'^* \cdot t) t + (\underline{\xi}'^* \cdot \underline{n}) \underline{n} = \underline{\varepsilon}^* t + \underline{\omega}^* \underline{n} \quad (6)$

$$7. \underline{\ddot{x}} = \left(\frac{d}{dh} \left(\frac{d}{ds} \omega_h \right) \right) \Big|_{h=0} = \left(\frac{d}{ds} \left(\frac{d}{dh} \omega_h \right) \right) \Big|_{h=0} = (\underline{\ddot{\omega}})^*$$

From virtual displacements/velocities to virtual work/power

Given $s \mapsto \underline{\xi}^*(s)$ field (virtual displacement/velocities; test field)

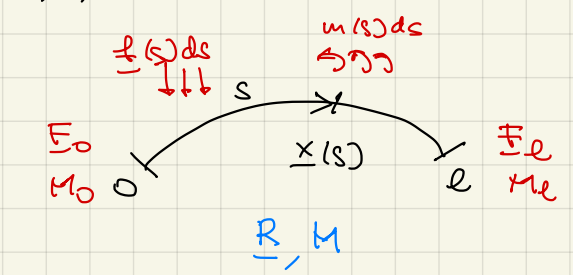
$\underline{\xi}^{*'} = \underline{\xi}^* \underline{t} + \underline{\omega}^* \underline{n}$, hence $\underline{\omega}^*(s) = \underline{\xi}^{*'} \cdot \underline{n}$, $\underline{\xi}^*(s) = \underline{\xi}^* \cdot \underline{t}$, $\underline{\dot{\xi}}^* = \underline{\omega}^*$ [kin. comp.]

Do the case without discontinuities of $\underline{\xi}^*$, or of M, N, T first

Assume $\underline{x}(s)$ is an equilibrium config. under

loads $\underline{F}_0, M_0, \underline{f}, m, \underline{F}_l, M_l$ (L)

with internal "forces" M, N, T



Equilibrium: $0 = \underline{R}' + \underline{f}$
 $0 = M' + T + m$

BC's: $\underline{R}(0) = -\underline{F}_0$, $\underline{R}(L) = \underline{F}_l$
 $M(0) = -M_0$, $M(L) = M_l$

$\cdot \underline{\xi}^*(s)$
 $\cdot \underline{\omega}^*(s)$ } sum and integrate $\int_0^L ds$

"test" equil. eqn's by multiplying by test functions (virtual work/power)

$0 = \int_0^L (\underline{R}' \cdot \underline{\xi}^* + \underline{f} \cdot \underline{\xi}^* + \underline{M}' \underline{\omega}^* + \underline{T} \underline{\omega}^* + m \underline{\omega}^*) ds = (*)$

$\int_0^L \underline{R}' \cdot \underline{\xi}^* ds = \int_0^L [(\underline{R} \cdot \underline{\xi}^*)' - \underline{R} \cdot \underline{\xi}^{*'}] ds = \underline{R}(L) \cdot \underline{\xi}^*(L) - \underline{R}(0) \cdot \underline{\xi}^*(0) - \int_0^L \underline{R} \cdot \underline{\xi}^{*'} ds$
 $= \underline{F}_l \cdot \underline{\xi}^*(L) + \underline{F}_0 \cdot \underline{\xi}^*(0) - \int_0^L (\underline{N} \underline{\xi}^* + \underline{T} \underline{\omega}^*) ds$

$\int_0^L M' \underline{\omega}^* ds = \int_0^L [(M \underline{\omega}^*)' - M \underline{\omega}^{*'}] ds = \underline{M}(L) \underline{\omega}^*(L) - \underline{M}(0) \underline{\omega}^*(0) - \int_0^L M \underline{\dot{\xi}}^* ds$ [$\underline{\omega}^* = \underline{\dot{\xi}}^*$]
 $= M_l \underline{\omega}^*(L) + M_0 \underline{\omega}^*(0)$

$(*) = \underline{F}_l \cdot \underline{\xi}^*(L) + M_l \underline{\omega}^*(L) + \underline{F}_0 \cdot \underline{\xi}^*(0) + M_0 \underline{\omega}^*(0)$ } (a)
 $+ \int_0^L (\underline{f} \cdot \underline{\xi}^* + m \underline{\omega}^*) ds$
 $- \int_0^L (\underline{N} \underline{\xi}^* + \underline{T} \underline{\omega}^*) ds$ } (b)

$+ \int_0^L \underline{T} \underline{\omega}^* ds - \int_0^L \underline{T} \underline{\omega}^* ds$

$= 0$ + virtual { displacement / velocity } field = test field

(a) = $P_{ext} [\underline{F}_l, M_l, \underline{F}_0, M_0, \underline{f}, m] (\underline{\xi}^*)$
L (loads)

external work/power

(b) = $P_{def} [M, N] (\underline{\xi}^*)$

internal or deformational work/power
 (it would be = 0 if $\underline{\xi}^*$ rigid)

PPV (a theorem!)

$\underline{x}(s)$ equilibrium configuration under loads $\underline{L}$
with internal "forces" M, N, T
(shears)

$$\Rightarrow P_{\text{ext}}[\underline{L}](\underline{\xi}^*) = P_{\text{def}}[M, N](\underline{\xi}^*) \quad \forall \underline{\xi}^*$$

admissible test field/perturbation

In fact, also the opposite implication $\Leftarrow$ is true.

This requires to know that $\int_0^L u(s) \varphi(s) ds = 0 \quad \forall \varphi \Rightarrow u(s) \equiv 0$

which φ 's? Fundamental lemma of Calculus of Variations

Using this fact, by following the argument (integration by parts) in the opposite direction we get

$$0 = \int_0^L (\underline{R}' \cdot \underline{\xi}^* + \underline{f} \cdot \underline{\xi}^* + M' \tilde{\omega}^* + T \tilde{\omega}^* + m \tilde{\omega}^*) ds \quad \forall \underline{\xi}^*$$

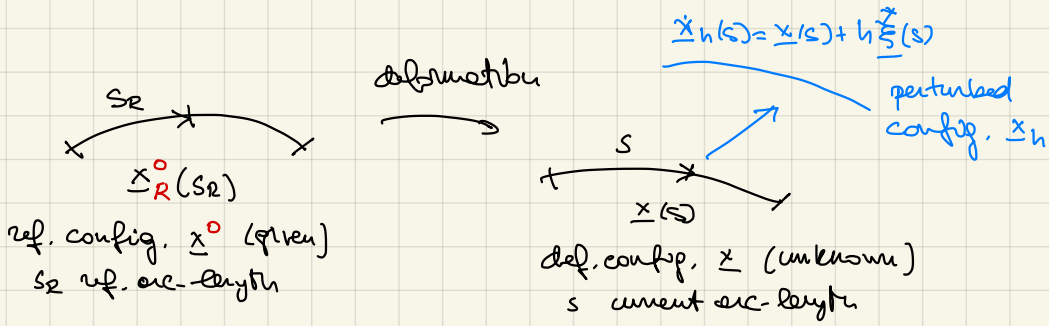
$(*) \quad \underline{R}' + \underline{f} = 0$
 $\Rightarrow M' + T + m = 0$

The integral identity $P_{\text{ext}}[\underline{L}](\underline{\xi}^*) = P_{\text{def}}[M, N](\underline{\xi}^*) \quad \forall \underline{\xi}^*$ admissible test field

is at the basis of the most wide-spread numerical procedure to solve the differential equations of equilibrium: weak form of differential equations, GALERKIN method (projection onto a finite-dimensional function space), Finite Element Method (FEM).

$(*)$ with some subtleties due to the fact that $\tilde{\omega}^* = \underline{\xi}^* \cdot \underline{n}$ is NOT indep. of $\underline{\xi}^*$,
see Merigo.

PPV in referential form



Re: $s = s(s_R)$

$$s(s_R) = \int_0^R \left| \frac{d}{ds_R} x(\sigma) \right| d\sigma$$

$$= \int_0^R (1 + E_R(\sigma)) d\sigma$$

↓
referential descript. of E

$$E_R(s_R) = E(s(s_R))$$

$$\frac{ds}{ds_R} = 1 + E_R(s_R)$$

Set $s = s(s_n)$ to get referential descriptions of $\underline{x}, \underline{x}_n$ and related quantities

$$\underline{x}(S(S_R)) = : \underline{x}_R(S_R)$$

$$\begin{aligned}\underline{x}_n(s(s_R)) &= \underline{x}(s(s_R)) + h \underline{\tilde{g}}(s(s_R)) \\ &= \underline{x}_R(s_R) + h \underline{\tilde{g}}_R(s_R) =: \underline{x}_{hR}(s_R)\end{aligned}$$

$$\underline{\text{PPV}} \quad \int_0^l \left(\frac{1}{2} \dot{\underline{z}}^T(s) \cdot \dot{\underline{z}}(s) + m(s) \dot{\underline{w}}^T(s) \dot{\underline{w}}(s) \right) ds = \int_0^l \left(M(s) \dot{\underline{z}}^T(s) \dot{\underline{z}}(s) + N(s) \dot{\underline{w}}^T(s) \dot{\underline{w}}(s) \right) ds$$

[for simplicity, no concentrated forces]

Change variables by setting $s = s(s_R)$, $ds = \frac{ds}{ds_R} ds_R = (1 + \epsilon_R) ds_R$ and get

$$\int_0^{\ell_R} (\underbrace{f_R(s_R)}_{\text{red wavy}} \cdot \underbrace{\tilde{m}_R^*(s_R)}_{\text{blue dashed}} + \underbrace{m_R(s_R)}_{\text{blue dashed}} \underbrace{\tilde{\omega}_R^*(s_R)}_{\text{red wavy}}) (1 + \varepsilon_R) ds_R = \int_0^{\ell_R} \underbrace{M_R(s_R)}_{\text{red wavy}} \underbrace{\tilde{\kappa}_R^*(s_R)}_{\text{blue dashed}} + \underbrace{N_R(s_R)}_{\text{red wavy}} \underbrace{\tilde{z}_R^*(s_R)}_{\text{blue dashed}} (1 + \varepsilon_R) ds_R$$

$$\Rightarrow \int_0^{\ell_R} (\underbrace{f_R}_{\text{red wavy}} \cdot \tilde{m}_R^* + m_R \tilde{\omega}_R^*) ds_R = \int_0^{\ell_R} (M_R (1 + \varepsilon_R) \tilde{\kappa}_R^* + N_R (1 + \varepsilon_R) \tilde{z}_R^*) ds_R \quad (*)$$

The structure of the deformation power density, e.g. $N_R (1 + \varepsilon_R)^* \dot{\varepsilon}_R$, is interesting. Why is this the right structure?

The basic answer is that

$N \vec{E} + M \vec{K}$ is the virtual deformation power density per unit current length

$$N(1+\varepsilon_R) \tilde{E} + M(1+\varepsilon_R) \tilde{K} \quad " \quad " \quad " \quad " \quad " \quad " \quad " \quad " \quad " \quad " \quad "$$

Other answers see next page

Answer 1 (SKIP)

(*) follows directly from differential equations of equilibrium in referential form

$$0 = \frac{d}{ds_R} R_R + \underline{f}_R = 0$$

$$0 = \frac{d}{ds_R} T_R + T_R (1 + \epsilon_R) + m_R = 0$$

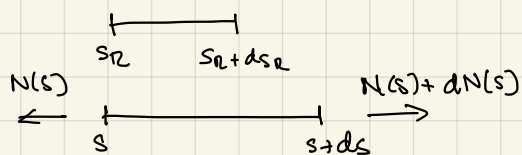
• $\underline{\tilde{\epsilon}}_R^*$
• $\underline{\tilde{\omega}}_R^*$
} sum and integrate $\int ds_R$ (Ext)

Answer 2 (Physical meaning)

ref. conf.

current conf.

perturbed conf.



To leading order, the work done is

$$N \cdot \underline{\tilde{\epsilon}} ds = N \underline{\tilde{\epsilon}}^* (1 + \epsilon_R) ds_R$$

$$\frac{\Delta l}{l} = \underline{\tilde{\epsilon}}^* \Rightarrow \Delta l = \underline{\tilde{\epsilon}}^* l$$

$$\underline{\tilde{\epsilon}}^* ds = \underline{\tilde{\epsilon}}^* (1 + \epsilon_R) ds_R$$

(increment of)

measured

The term $\underline{\tilde{\epsilon}}^* (1 + \epsilon_R)$ represents, in fact, the virtual extensional strain with respect to the reference configuration $\underline{x}^0$ associated with the perturbation $\underline{x} \mapsto \underline{x}_h$: see later

Remark

Up to now, PPV entirely general: no const. eq. or M, N ; no hyp. or loads being conservative.

when forces are conservative (e.g., centrifugal force) and, in particular, constant (e.g., gravity)

and const. eq. derive from internal elastic energy

$$N = EA \underline{\epsilon} \rightarrow \exists W(\underline{\epsilon}, \kappa) \text{ s.t. } N = \frac{\partial W}{\partial \underline{\epsilon}}(\underline{\epsilon}, \kappa)$$

$$M = EI \kappa$$

$$M = \frac{\partial W}{\partial \kappa}(\underline{\epsilon}, \kappa)$$

$$W(\underline{\epsilon}, \kappa) = \frac{1}{2} EA \underline{\epsilon}^2 + \frac{1}{2} EI \kappa^2$$

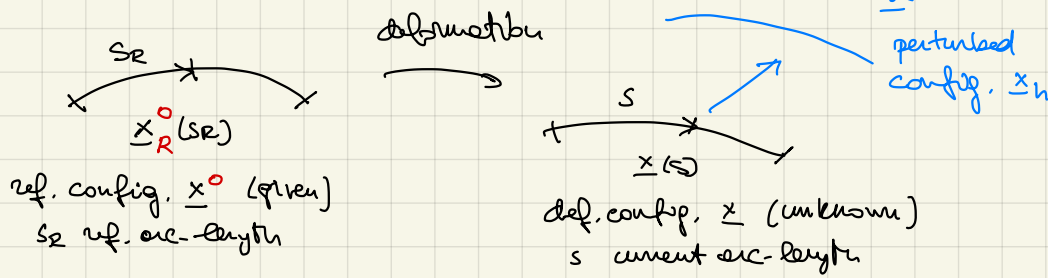
$\underline{\epsilon}, \kappa$ measured with respect to stress-free initial configuration $\underline{x}^0$

[ext. forces admit a potential]

[int. forces admit a potential]

} $\rightarrow$ PPV becomes principle of stationary / minimal potential energy

Towards principle of min. pot. energy



More formally, consider referential description of extensional and bending strain and rotation in perturbed configuration $\underline{x}_h$ measured with respect to the reference configuration $\underline{x}^0$

$$1 + \varepsilon_{hr}^0 := \left| \frac{d}{ds_R} \underline{x}_h(s(s_R)) \right| = \left| \frac{d}{ds} \underline{x}_h(s(s_R)) \frac{ds}{ds_R} \right| = \left| \frac{d}{ds} \underline{x}_h(s(s_R)) \right| (1 + \varepsilon_R) = (1 + \varepsilon_{hr}) (1 + \varepsilon_R)$$

$1 + \varepsilon_{hr}^0 :=$

$\frac{ds}{ds_R} = 1 + \varepsilon_R \geq 0$

Heuristically: $1 + \varepsilon_{hr}^0 = \frac{ds_h}{ds_R} = \frac{ds_h}{ds} \frac{ds}{ds_R} = (1 + \varepsilon_{hr})(1 + \varepsilon_R)$

extension in $\underline{x}^0 \rightarrow \underline{x}$

extension in $\underline{x} \rightarrow \underline{x}_h$

both expressed as functions of s_R

Similarly

$$\omega_{hr}^0 := \alpha_{hr} - \alpha_R^0 \quad \pm \alpha_R$$

$$= (\alpha_{hr} - \alpha_R) + (\alpha_R - \alpha_R^0) = \omega_h(s(s_R)) + \omega_R(s_R)$$

$\omega_{hr} = \omega_h(s(s_R))$

$\omega_R(s_R)$ rotation in $\underline{x}^0 \rightarrow \underline{x}$

rotation in $\underline{x} \rightarrow \underline{x}_h$

$$k_{hr}^0 := \frac{d}{ds_R} \omega_{hr}^0 = \frac{d}{ds} \omega_h(s(s_R)) \frac{ds}{ds_R} + \frac{d}{ds_R} \omega_R(s_R)$$

ω_{hr}'

$1 + \varepsilon_R$

k_R

$$= k_{hr} (1 + \varepsilon_R) + k_R$$

Therefore

$$\left. \frac{d}{dh} \varepsilon_{hr}^0 \right|_{h=0} = \left. \frac{d}{dh} \varepsilon_{hr} \right|_{h=0} (1 + \varepsilon_R) = \bar{\varepsilon}_R (1 + \varepsilon_R) \quad \left[= \bar{\varepsilon}_R^0 \right] \quad (1)$$

$$\left. \frac{d}{dh} k_{hr}^0 \right|_{h=0} = \left. \frac{d}{dh} k_{hr} \right|_{h=0} (1 + \varepsilon_R) = \bar{k}_R (1 + \varepsilon_R) \quad \left[= \bar{k}_R^0 \right] \quad (2)$$

Introduce now constitutive equations, and their descriptions through elastic potential

$$W_{el}(\varepsilon_R^0, \kappa_R^0) \quad \text{s.t.}$$

$$N_R(\varepsilon_R^0, \kappa_R^0) = \frac{\partial}{\partial \varepsilon_R^0} W_{el}(\varepsilon_R^0, \kappa_R^0)$$

$$M_R(\varepsilon_R^0, \kappa_R^0) = \frac{\partial}{\partial \kappa_R^0} W_{el}(\varepsilon_R^0, \kappa_R^0)$$

In our linear case, $W_{el} = \frac{1}{2} EA (\varepsilon_R^0)^2 + \frac{1}{2} EI (\kappa_R^0)^2$

$$\Rightarrow N_R = EA \varepsilon_R^0, \quad M_R = EI \kappa_R^0, \quad \text{both } = 0 \text{ in the ref. conf. } \underline{x}^0$$

$$\text{where } \varepsilon_R^0 = 0, \quad \kappa_R^0 = 0$$

Define:

$$V_{el}(\underline{x}_h) := \int_0^{l_R} W_{el}(\varepsilon_{hR}, \kappa_{hR}) ds_R$$

[change of elastic energy going from $\underline{x}^0$ to $\underline{x}_h$]

$$W_{ext}[L](\underline{x}_h) := \text{work done by constant (conservative) forces going from } \underline{x}^0 \text{ to } \underline{x}_h$$

$$= \underline{F}_{sRi} \cdot (\underline{x}_{hR}(s_{Ri}) - \underline{x}_R^0(s_{Ri})) \quad [\text{e.g. constant force at } s(s_{Ri})]$$

$$= \underline{F}_{sRi} \cdot \left(\underbrace{\underline{x}_{hR}(s_{Ri}) - \underline{x}_R(s_{Ri})}_{h \underline{\xi}_R^*(s_{Ri})} + \underbrace{\underline{x}_R(s_{Ri}) - \underline{x}_R^0(s_{Ri})}_{\underline{\xi}_R(s_{Ri})} \right) = \underline{F}_{sRi} \cdot h \underline{\xi}_R^*(s_{Ri}) + \underline{F}_{sRi} \cdot \underline{\xi}_R(s_{Ri})$$

$$V(\underline{x}_h) := V_{el}(\underline{x}_h) - W_{ext}[L](\underline{x}_h)$$

total potential energy in $\underline{x}_h$

Compute $\left. \frac{d}{dh} V \right|_{h=0} = \left. \frac{d}{dh} V_{el} \right|_{h=0} - \left. \frac{d}{dh} W_{ext} \right|_{h=0}$

$$\begin{aligned} \bullet \left. \frac{d}{dh} V_{el} \right|_{h=0} &= \int_0^{l_R} \left(\frac{\partial W_{el}}{\partial \varepsilon_R^0}(\varepsilon_{hR}, \kappa_{hR}) \underbrace{\left. \frac{d}{dh} \varepsilon_{hR} \right|_{h=0}}_{(1)} + \frac{\partial W_{el}}{\partial \kappa_R^0}(\varepsilon_{hR}, \kappa_{hR}) \underbrace{\left. \frac{d}{dh} \kappa_{hR} \right|_{h=0}}_{(2)} \right) ds_R \\ &= \int_0^{l_R} \left(N_R(\varepsilon_R^0, \kappa_R^0) \underline{\xi}_R^*(1+\varepsilon_R) + M_R(\varepsilon_R^0, \kappa_R^0) \underline{\kappa}_R^*(1+\varepsilon_R) \right) ds_R \\ &= P_{def}[N_R, M_R](\underline{\xi}_R^*) \end{aligned}$$

$$\bullet \left. \frac{d}{dh} W_{ext} \right|_{h=0} = \underline{F}_{sRi} \cdot \underline{\xi}_R^*(s_{Ri}) = P_{ext}[L](\underline{\xi}_R^*)$$

$\Rightarrow$ Conclude that $\left. \frac{d}{dh} V \right|_{h=0} = 0 \quad \forall \underline{\xi}_R^*$ is PPV in referential form.

Summary

We have shown that

$$\left. \frac{d}{dn} v(x+h\underline{\xi}) \right|_{n=0} = 0 \quad \forall \underline{\xi}^* \quad \text{total potential energy stationary / minimal}$$

is equivalent to PPV in differential form

$$P_{\text{ext}}[L](\underline{\xi}_R^*) = P_{\text{def}}[M_R, N_R](\underline{\xi}_R^*) \quad \forall \underline{\xi}_R^*$$

This will be our tool to write down equilibrium equations in the large deformation regime.

Recap on PPV

- PPV for rigid systems

$$\underline{\dot{\mathbf{g}}}(L) = \underline{0}$$

$$\underline{\mathbf{m}}_0(L) = \underline{0}$$

$$\Leftrightarrow \mathcal{P}_{\text{ext}}[L](\underline{\dot{\mathbf{g}}}^*) = 0 \quad \forall \underline{\dot{\mathbf{g}}}^* \text{ test}$$

(virtual rigid velocity)

- PPV for deformable systems

$$\underline{\mathbf{R}'} + \underline{\mathbf{f}} = \underline{0}$$

$$\underline{\mathbf{M}'} + \underline{\mathbf{T}} + \underline{\mathbf{m}} = \underline{0}$$

$$\Leftrightarrow$$

$$\mathcal{P}_{\text{ext}}[L](\underline{\dot{\mathbf{g}}}^*) = \mathcal{P}_{\text{int}}[M, N](\underline{\dot{\mathbf{g}}}^*) \quad \forall \underline{\dot{\mathbf{g}}}^*$$

(internal forces balance)
external loads

(external virtual power balances)
internal/dif. virtual power

$$\mathcal{P}_{\text{int}}[M, N](\underline{\dot{\mathbf{g}}}^*) = \int_0^L (\underline{\mathbf{M}} \underline{\dot{\mathbf{K}}} + \underline{\mathbf{N}} \underline{\dot{\mathbf{E}}}) ds = \int_0^{L_R} (\underline{\mathbf{M}}_R \underline{\dot{\mathbf{K}}}_R + \underline{\mathbf{N}}_R \underline{\dot{\mathbf{E}}}_R)(1 + \epsilon_R) dS_R$$

- Deformation/internal power density

$$(\underline{\mathbf{M}} \underline{\dot{\mathbf{K}}} + \underline{\mathbf{N}} \underline{\dot{\mathbf{E}}}) ds$$

per unit current length

$$(\underline{\mathbf{M}}_R \underline{\dot{\mathbf{K}}}_R + \underline{\mathbf{N}}_R \underline{\dot{\mathbf{E}}}_R)(1 + \epsilon_R) dS_R$$

per unit reference density

$$\underline{\dot{\mathbf{E}}}_R(1 + \epsilon_R) = \underline{\dot{\mathbf{E}}}_R^0$$

The term $\underline{\dot{\mathbf{E}}}_R(1 + \epsilon_R)$ represents, in fact, the virtual rate of increment of extensional strain, measured with respect to the reference configuration, associated with the perturbation $\underline{x} \rightarrow \underline{x}_h$.

Similar interpretation for $\underline{\dot{\mathbf{K}}}_R(1 + \epsilon_R) = \underline{\dot{\mathbf{K}}}_R^0$

- with conservative loads and hyperelastic constitutive equations
PPV becomes principle of minimal/stationary potential energy.

Extraneous: presence of faces and constraints at internal points, elastic constraints

see: Complements to lecture 18